

Data Flow Diagram

Date	4 July 2026
Team ID	
Project Name	Comprehensive measure of well-being
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

DFD Components Mapping

1. External Entities (Ovals)

- **User:** Government policymaker or development analyst who interacts with the application interface.

2. Processes (Numbered Rectangles)

- **1 Request Handling / Input Validation:** Extracts user inputs from the web page form, validates formatting constraints, and ensures numbers fall within logical ranges.
- **2 Analysis / Index Prediction:** Ingests the validated metrics array, runs them through the pre-compiled Machine Learning intelligence calculations, and computes the well-being value bounded between 0.0 and 1.0.

3. Data Stores (Solid Rectangles)

- **Model Serialization File (HDI.pkl):** The persistent background store holding pre-trained linear regression mathematical weights.

4. Data Flows (Labeled Arrows)

- "Socio-economic parameters" → From User To Process 1 (Input Validation)
- "Validated parameters array" → From Process 1 To Process 2 (Index Prediction)
- "Model coefficients & intercept data" → From Data Store (HDI.pkl) To Process 2
- "Calculated HDI value (0.0 - 1.0)" → From Process 2 To User

Diagram

